

18.75
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22P0030

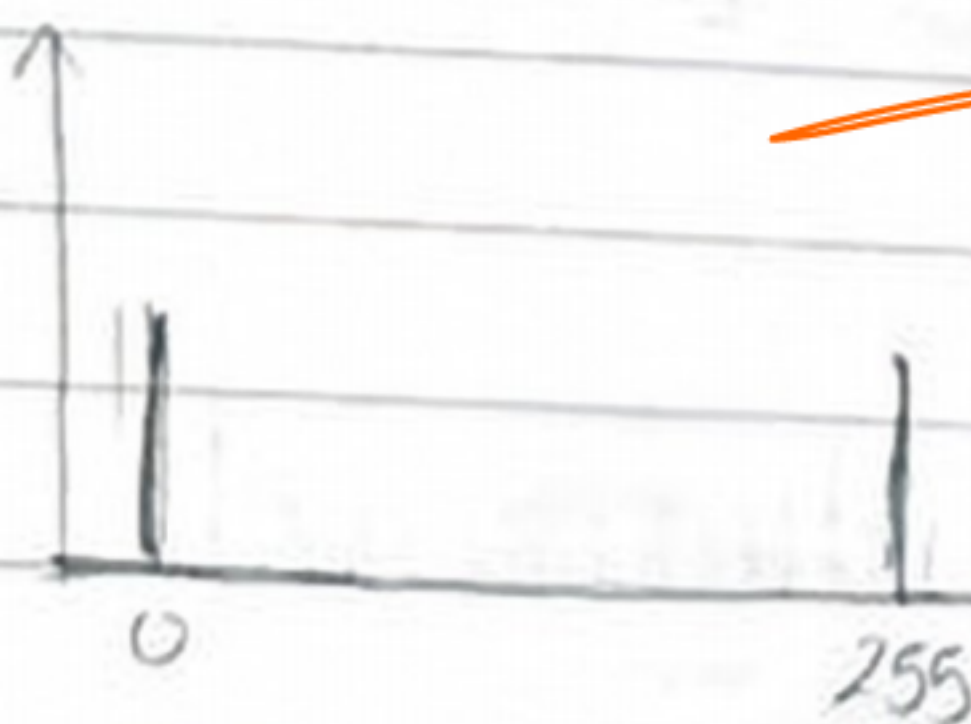
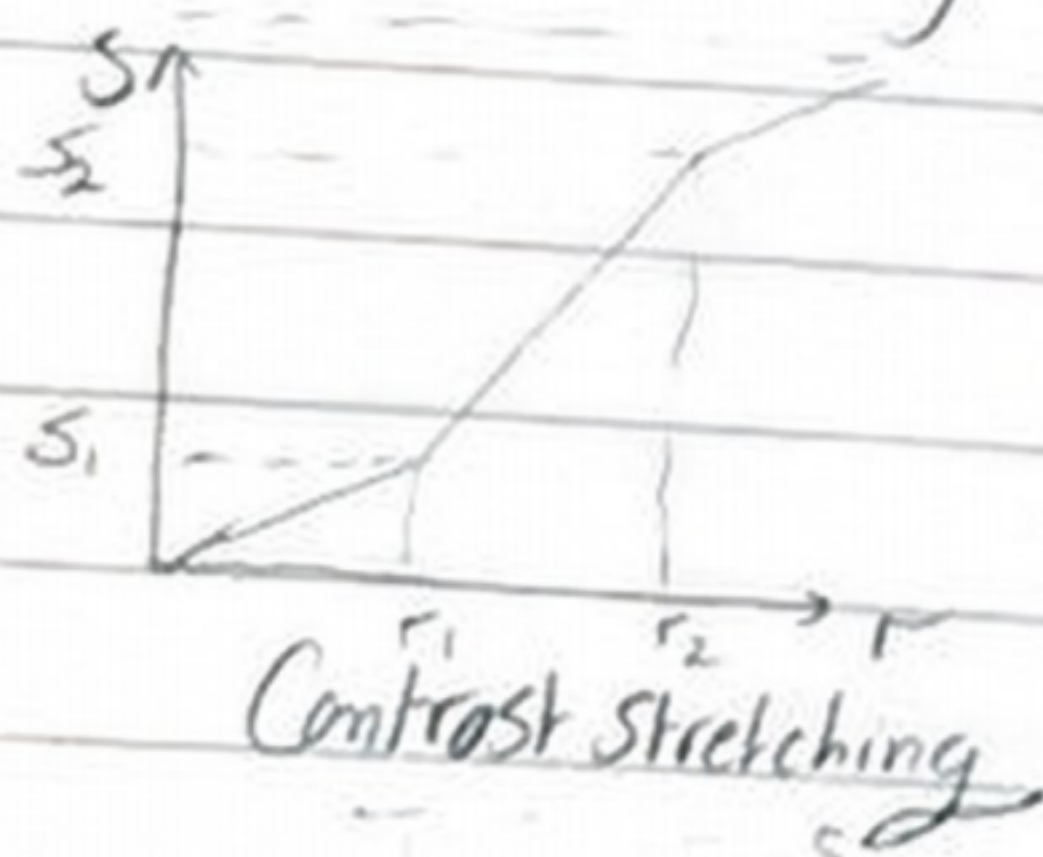
ECE359 Mid Term
Spring 2026

Question 1]

- intensity values of the given histogram

0, 50, 100, 150, 200, 250
handwriting shading Background

- we will use Contrast Stretching Technique for enhancement & outputting a high Contrast and Thresholding to remove shading of background



it should have high value at 0 and at 255
Handwriting Background

Excellent! Φ

Question 4

$A(3,3)$ $B(7,3)$ $C(7,7)$ $D(3,7)$

$A'(6.5,3.4)$ $B'(12.5,3.4)$ $C'(12.5,6.6)$ $D'(6.5,6.6)$

$$x \rightarrow 6.5 - 3 = 3.5$$

$$12.5 - 7 = 5.5$$

Scaling by $1/7$

$\frac{AB'}{A'B'}$ not $\frac{A'A'}{B'B'}$

$$y \rightarrow 3.4 - 3 = 0.4$$

$$6.6 - 7 = -0.4$$

Scaling by -1

$$x' = \frac{11}{7}x + tx \rightarrow 6.5 = \frac{11}{7} \times 3 + tx \quad tx = \frac{25}{14}$$

$$y' = -y + ty \rightarrow 3.4 = -3 + ty \quad ty = 6.4$$

$$A = \begin{bmatrix} 11/7 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 25/14 \\ 0 & 1 & 6.4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 11/7 & 0 & 275/98 \\ 0 & -1 & -6.4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\frac{25}{3}$$

Steps are correct
good work

Question 3]

8/8 Excellent

1. GD-TA-H2 ✓ There is a slight brightening in the image so a slight shift in histogram
2. GB-TC-H4 ✓ graph B represents darkening Transformation which is TC image and its histogram is H4 as it is shifted left
3. GC-TD-H1 ✓ graph C represents brightening Transformation which is TD image & its histogram is H1 as it is shifted right
4. GF-TE-H5 ✓ Thresholding appears which means only two intensities available black or white
5. GE-TB-H3 ✓ inverse Transformation where bright turns dark and dark turns bright
6. Graph A is not used as we don't have the same image and the same histogram & graph A represents identity
 ✗

Question 4]

r_{kq}	n_k	$p(r_k)$	s_k	n_q	$p_z(z_q)$	v_q	$R \rightarrow Z$	z
0	1	1/32	0.2 ≈ 0	0	0	0	0 → 0	✓
1	4	4/32	1.0 ≈ 1	1	0	0	1 → 2	✓
2	8	8/32	2.84 ≈ 2	4	4/32	0.875 ≈ 1	2 → 3	✗ 4
3	5	5/32	3.9 ≈ 4	4	4/32	1.75 ≈ 2	3 → 5	✓
4	2	2/32	4.3 ≈ 4	4	4/32	2.625 ≈ 3	4 → 5	✓
5	2	2/32	4.8 ≈ 5	4	4/32	3.5 ≈ 4	5 → 6	✓
6	5	5/32	5.9 ≈ 6	5	5/32	4.59 ≈ 5	6 → 6	✓
7	5	5/32	7	11	11/32	7	7 → 7	✓
	Σ 32							↑

image after Mapping

0 → will exist 1 time

2 → 4 Times

3 → 8 times

5 → 7 times

6 → 7 times

7 → 5 times

We cannot draw exact Location as we don't have Spatial

Coordinates ✗

we have the image !

-0.5

~~4.25~~
~~5~~